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UTS MTK SAINS DATA

Parameter Personalisasi

1. menentukan N
   $N = 31$
2. menentukan a
   $a = \text{floor}\left(\frac{31}{10}\right) = 3$
3. menentukan b
   $b = 31 \bmod 10 = 1$
4. hitung k
   $k = a + b + 1$
   $k = 3 + 1 + 1 = 5$
5. hitung θ
   $\theta_0 = (a \times 30) + (b \times 5)$
   $\theta_0 = (3 \times 30) + (1 \times 5) = 90 + 5 = 95^\circ$
6. hitung α
   $\alpha = 0.001 \times (b + 1)$
   $\alpha = 0.001 \times (1 + 1) = 0{,}002$

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Parameter

$a = 3, b = 1, N = 31$

(a) Matriks A dan determinan matriks kunci

$$A = \begin{bmatrix} a+1 & b & 2 \\ 1 & a+2 & b \\ b & 1 & a+3 \end{bmatrix} = \begin{bmatrix} 4 & 1 & 2 \\ 1 & 5 & 1 \\ 1 & 1 & 6 \end{bmatrix}$$

Det baris 1

$$\det(A) = 4\begin{vmatrix} 5 & 1 \\ 1 & 6 \end{vmatrix} - 1\begin{vmatrix} 1 & 1 \\ 1 & 6 \end{vmatrix} + 2\begin{vmatrix} 1 & 5 \\ 1 & 1 \end{vmatrix}$$

minor $= 4(30-1) = 4(29) = 116$

$= -1(6-1) = -1(5) = -5$

$= +2(1-5) = 2(-4) = 8$

$\det(A) = 116 - 5 - 8 = 103$

(b) Matriks minor

$$M \begin{bmatrix} 29 & 5 & -4 \\ 4 & 22 & 3 \\ -9 & 2 & 19 \end{bmatrix}$$

(c) Kofaktor, Adj, dan invers

Kofaktor

$$C = \begin{bmatrix} 29 & -5 & -4 \\ -4 & 22 & -3 \\ -9 & -2 & 19 \end{bmatrix}$$

Adjoin

$$Adj(A) \begin{bmatrix} 29 & -4 & -9 \\ -5 & 22 & -2 \\ -4 & -3 & 19 \end{bmatrix}$$

Invers

$$A^{-1} = \frac{1}{103}\begin{bmatrix} 29 & -4 & -9 \\ -5 & 22 & -2 \\ -4 & -3 & 19 \end{bmatrix}$$

(e) Enkripsi dan dekripsi

matriks P

$$P \begin{bmatrix} 120 & 81 & 200 \\ 80 & 5 & 41 \\ 61 & 90 & 170 \end{bmatrix}$$

Enkripsi $E = A \cdot P$

$$E \begin{bmatrix} 682 & 509 & 1141 \\ 581 & 196 & 575 \\ 566 & 626 & 1261 \end{bmatrix}$$

Dekripsi $P' = A^{-1} \cdot E$

$$P' = \begin{bmatrix} 120 & 81 & 200 \\ 80 & 5 & 41 \\ 61 & 90 & 170 \end{bmatrix}$$

$P' = P$

Jika $\det(A) = 0$ maka

- matriks tdk punya invers
- Dekripsi tidak bisa
- Enkripsi juga gagal

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